

Torsion and Tamagawa numbers over function fields

Modular Curves and Applications of AI to Number Theory

David Kurniadi Angdinata (with Mentzelos Melistas)

University of East Anglia

Thursday, 17 September 2026

The Birch and Swinnerton-Dyer conjecture

Let E be an elliptic curve over a global field K .

Tate 1966 conjectured that $\text{ord}_{s=1} L(E, s) = \text{rk}(E)$, and that

$$L^*(E) = \frac{\Omega(E) \cdot \text{Reg}(E) \cdot \#\text{III}(E) \cdot c(E)}{\mu(\mathbb{A}_K/K) \cdot \#\text{tor}(E)^2}.$$

Here, $\text{tor}(E)$ is the **torsion subgroup**, and the **Tamagawa number** is

$$c(E) := \prod_v c_v(E) := \prod_v [E(K_v) : E_0(K_v)],$$

where $c_v(E)$ counts the k_v -rational components of its Néron model at v .

Are there any cancellations between $\#\text{tor}(E)^2$ and $c(E)$?

Known divisibilities

Let E be an elliptic curve over a number field K .

Theorem (Lorenzini 2011)

If $K = \mathbb{Q}$ and $\text{tor}(E) \cong T$, then c divides $c(E)$.

T	c	exceptions
$C_4, C_2 \oplus C_4$	2	$X_1(15), 15a7, 17a4$
$C_6, C_2 \oplus C_6$	6	$X_1(14), 14a6, 20a2$
$C_5, C_7, C_8, C_9,$ $C_{10}, C_{12}, C_2 \oplus C_8$	$\#T$	$X_1(11)$

If $K = \mathbb{Q}(\sqrt{d})$ and E has a point of order $N = 11, 13$, then N divides $c(E)$. In general, if E has a point of order $N = 7, 9$, then N divides $c(E)$, up to at most $2^{9(\text{rk } \mathcal{O}_K^\times + 1)}$ exceptions.

Proof strategy

Let P be a point of order $N \geq 4$. Then E has a **Tate normal form**

$$\mathcal{T}(a, b) : y^2 + (1 - a)xy - by = x^3 - bx^2, \quad a, b \in K,$$

with $P = (0, 0)$, and $[N](0, 0) = O$ gives an affine model of $X_1(N)$.

If $v(a), v(b) > 0$ for some place v of K , then the reduction of E at v is $y^2 + xy = x^3$, which is split multiplicative with $c_v(E) = v(\Delta)$.

Example ($K = \mathbb{Q}$ and $N = 5$)

The condition $2P = -3P$ forces $a = b =: f$ with $\Delta = f^5(f^2 - 11f - 1)$.

If $v(f) \neq 0$ for some place v of $\mathbb{Q}$, then assuming $v(f) > 0$ gives $c_v(E) = 5v(f)$. Otherwise, $f \in \mathbb{Z}^\times = \{\pm 1\}$ forces $E = X_1(11)$.

Known divisibilities

Let K be the function field of a nice curve C of genus g over $k = \mathbb{F}_q$, and let E be a non-isotrivial elliptic curve over K of height $h = \deg \omega_E > 0$.

Then $\# \operatorname{tor}(E)$ divides $c(E)$ easily, by considering the map

$$\operatorname{tor}(E) \hookrightarrow E(K) \hookrightarrow \prod_v E(K_v) \rightarrow \prod_v E(K_v)/E_0(K_v),$$

which is injective by Shioda's height formula

$$\langle P, P \rangle = 2h + 2(P \cdot O) - \sum_v \operatorname{contr}_v(P), \quad P \in E(K).$$

Ulmer 2014 also proved that $\# \operatorname{tor}(E)^2$ divides $\operatorname{Reg}(E) \cdot D(E)$, where $D(E)$ is the absolute discriminant of the trivial lattice over k of the minimal proper regular model of E , which is related to $c(E)$.

New divisibilities

Theorem (A.–Melistas 2026)

If $K = k(t)$ with $\text{char}(k) \neq 2, 3$, then $\# \text{tor}(E)^2$ divides $c(E)$, except when $\# \text{tor}(E) = 2, 3, 4$.

Since $\Omega(E) = q^{-h}$ and $\mu(\mathbb{A}_K/K) = q^{-1}$,

$$\frac{q^h}{\text{Reg}(E)} \cdot L^*(E) \stackrel{\text{BSD}}{=} \frac{q \cdot \# \text{III}(E) \cdot c(E)}{\# \text{tor}(E)^2} \in \mathbb{Z}[\tfrac{1}{6}].$$

Theorem (A.–Melistas 2026)

A point of prime order $7 \leq N \leq 101$ with $N \nmid q$ forces N^2 to divide $c(E)$.

This should hold for all primes $N \geq 5$, including $N \mid q$.

Sharp divisibilities

T	$\text{char}(k)$	c
C_2	any	2
C_3	any	3
C_4	$\neq 2$	2^3
	2	2^2
C_5	any	5^2
C_6	$\neq 2, 3$	6^2
	2	$2 \cdot 6$
	3	$3 \cdot 6$
C_7	any	7^3
C_8	$\neq 2$	8^3
	2	8^2
C_9	$\neq 3$	$3 \cdot 9^3$
	3	9^3
C_{10}	$\neq 2, 5$	$5 \cdot 10^3$
	2	$2 \cdot 10^2$
	5	$5^2 \cdot 10^2$
C_{12}	$\neq 2, 3$	12^4
	2	$2^2 \cdot 12^2$
	3	$3 \cdot 6 \cdot 12^2$

T	$\text{char}(k)$	c
$C_2 \oplus C_2$	$\neq 2$	2^3
$C_3 \oplus C_3$	$\neq 3$	3^4
$C_4 \oplus C_2$	$\neq 2$	$4^2 \cdot 2^2$
$C_6 \oplus C_2$	$\neq 2, 3$	$2^3 \cdot 6^3$
	3	6^3
$C_8 \oplus C_2$	$\neq 2$	2^{16}
$C_4 \oplus C_4$	$\neq 2$	4^6
$C_6 \oplus C_3$	$\neq 2, 3$	$3^4 \cdot 6^4$
	2	6^4
$C_5 \oplus C_5$	$\neq 5$	5^{12}
C_{11}	11	11^5
C_{14}	2	$2 \cdot 14^3$
	7	$7^3 \cdot 14^3$
C_{15}	3	$3 \cdot 15^4$
	5	$5^2 \cdot 15^4$
C_{18}	2	$2^2 \cdot 3 \cdot 18^3$
$C_{10} \oplus C_2$	5	10^6
$C_{10} \oplus C_5$	2	10^{12}
$C_{12} \oplus C_2$	3	$6^2 \cdot 12^4$

Proof strategy

If P is a point of order $N \geq 4$, then the Tate normal form gives a morphism $f : C \rightarrow X_1(N)$, which is surjective since $j \notin k$.

Example ($K = k(t)$ and $\text{tor}(E) \cong C_5$)

The morphism $f : \mathbb{P}_k^1 \rightarrow X_1(5) \cong \mathbb{P}_f^1$ has a zero v_1 and a pole v_2 , lying over the cusps 0 and ∞ of $X_1(5)$, which are distinct places of K .

As before, $c_{v_1}(E) = v_1(\Delta) = 5v_1(f)$ at v_1 .

On the other hand, the change of variables $(x, y) \mapsto (x/f^2, y/f^3)$ gives

$$y^2 - (1 - f^{-1})xy - f^{-2}y = x^3 - f^{-1}x^2, \quad \Delta = f^{-5} - 11f^{-6} - f^{-7},$$

whose reduction at v_2 is $y^2 - xy = x^3$, which is also split multiplicative with $c_{v_2}(E) = -5v_2(f)$.

McDonald parameterisations

Theorem (Cox–Parry 1980, McDonald 2018)

Let E be a non-isotrivial elliptic curve over $k(t)$ with a point of order $N \geq 4$. Then $\text{tor}(E) \cong T$ with an explicit Tate normal form.

$\text{char}(k)$	T
any	$C_4, C_5, C_6, C_7, C_8, C_9, C_{10}, C_{12}$
$\neq 2$	$C_4 \oplus C_2, C_6 \oplus C_2, C_8 \oplus C_2, C_4 \oplus C_4$
$\neq 3$	$C_6 \oplus C_3$
$\neq 5$	$C_5 \oplus C_5$
2	$C_{14}, C_{18}, C_{10} \oplus C_5$
3	$C_{15}, C_{12} \oplus C_2$
5	$C_{15}, C_{10} \oplus C_2$
7	C_{14}
11	C_{11}

Lean formalisation

As part of Axiom Math, I manually formalised Tate's algorithm over perfect residue fields, which *defines* the Tamagawa number.

This definition works over any commutative ring equipped with a reduction datum and an additive valuation that are compatible, including $\mathbb{Z}_p$ and $k[[t]]$, but also $\mathbb{Z}$ and $k[t]$ upon choosing a uniformiser.

This definition is good for proofs, but also for kernel computation given a strategy to compute p -th roots and roots of cubics when $\text{char}(k) = p$.

Using GPT-5.6 Sol, I autoformalised the divisibility results and the sharp examples over a few weeks, with refactoring still ongoing with Fable 5.1.

The total lines of code is approximately 3,000 lines for Tate's algorithm, 30,000 lines for divisibility results, and 30,000 lines for sharp examples.

Reference errors

GPT-6 Astra detected minor mistakes in the literature.

Ulmer 2019 claims that $\# \operatorname{tor}(E)^2$ divides $\operatorname{Reg}(E) \cdot c(E)$, based on a claimed proof that $c(E) = D(E)$ by Berger et al 2020. However, the elliptic curve E over $\mathbb{F}_3(t)$ given by $y^2 + xy + (t^3 - t + 1)y = x^3$ has

$$\frac{\operatorname{Reg}(E) \cdot c(E)}{\# \operatorname{tor}(E)^2} = \frac{1}{3}.$$

McDonald 2018 claims that $\operatorname{tor}(E) \cong C_{12} \oplus C_2$ when $\operatorname{char}(k) = 3$ can only occur when $\zeta_4 \in k$. However, the elliptic curve E over $\mathbb{F}_3(t)$ given by

$$y^2 + xy + \frac{t^3(t^2 - 1)^3}{(t^2 + 1)^6}y = x^3 + \frac{t^3(t^2 - 1)^3}{(t^2 + 1)^6}x^2$$

has $\operatorname{tor}(E) \cong C_{12} \oplus C_2$, but $\zeta_4 \notin \mathbb{F}_3$.

Future work

Observe that $5^2 \mid c(E)$ when $\text{tor}(E) \cong C_5$, but also $7^3 \mid c(E)$ when $\text{tor}(E) \cong C_7$ and $11^5 \mid c(E)$ when $\text{tor}(E) \cong C_{11}$.

Conjecture

A point of prime order $N \geq 5$ forces $N^{(N-1)/2} \mid c(E)$.

Conjecture

Independently of the Birch and Swinnerton-Dyer conjecture,

$$\frac{q^{h+g-1}}{\text{Reg}(E)} \cdot L^*(E) \in \mathbb{Z}[\tfrac{1}{6}], \quad \frac{c(E)}{\#\text{tor}(E)^2} \in \mathbb{Z}[\tfrac{1}{6}].$$

GPT-6 Astra claims $q^{h+g-1} \cdot L^*(E) \in \mathbb{Z}$ by calculating Newton slopes of $L(E, T)$ as the characteristic polynomial of Frobenius acting on the crystalline cohomology of the surface associated to E .